

Minsoo Ryu

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Seoul, Republic of Korea

BIO

Minsoo Ryu received an M.S. degree in Cybersecurity from the School of Cybersecurity, Korea University, in 2022. He received a B.E. degree in Computer and Information Security from Sejong University, in 2020. He is currently a Research Engineer (Senior Level / R&D Lead) at Bluebird, Inc. His primary expertise and recent research interests include Android, Automotive, and Robotics.

TECHNICAL SKILLS

Programming Language: C, C++, Python, Kotlin, Java

OS: Linux, Android, Embedded Linux, RTOS (Real-Time OS), AGL (Automotive Grade Linux)

Protocols: Apple VAS, ISO-DEP, APDU

Debugging & Profiling: Low-Level Kernel/Protocol Debugging, Root-Cause Analysis, Troubleshooting, Memory Leak Optimization (OOM)

System & Architecture: IPC (AIDL), Multi-threading, Memory Management

Security: Reversing, Debugging, Vulnerability Assessment

EDUCATION

2020.09-2022.08, M.S., Cybersecurity, Korea University (Seoul, Korea)

2016.03-2020.08, B.E., Computer and Information Security, Sejong University (Seoul, Korea)

2018.07-2019.03, Next-generation security leader training program, KITRI (Seoul, Korea)

WORKING EXPERIENCE

2026.03-Present (4 months ~), Research Engineer at Bluebird, Inc.

2024.03-2026.02 (2 years), Assistant Research Engineer at Bluebird, Inc.

2023.06-2024.02 (9 months), Engineer at Bluebird, Inc.

2020.09-2022.08 (2 years), Graduate Research Assistant at HCRL (Hacking and Countermeasure Research Lab)

2020.01-2020.03 (3 months), Researcher at Pizzlysoft

2019.04-2019.12 (9 months), Penetration Tester at Posco ICT

RESEARCH and PROFESSIONAL EXPERIENCE

[PJ18] 2026.05-2026.06 (2 months), Development of Newly Applied VAN (NICE)

[PJ17] 2025.11-Present (8 months ~), Architecture Design and Development of Apple VAS (Tech Lead)

[PJ16] 2025.02-Present (6 months ~), Development of Management Android Application

[PJ15] 2025.04-2025.07 (4 months), Development of Newly Applied VAN (KCP)

[PJ14] 2025.03-2025.03 (1 month), Low-Level Debugging and Root-Cause Analysis for KR Cash IC Sub-System Issues

[PJ13] 2025.01-2025.02 (2 months), Development of Newly Applied VAN (KSNet)

[PJ12] 2024.07-2025.03 (9 months), Development of BMT Applications

[PJ11] 2024.06-2024.11 (6 months), Management of Outsourcing Development Project

[PJ10] 2023.07-2024.04 (10 months), Development of Mastercard Contactless Kernel

[PJ9] 2022.09-2022.11 (3 months), Cybersecurity AI/Big Data Challenge 2022: AI-based Malware Threat Classification

[PJ8] 2022.02-2022.02 (1 month), Development of an APK Crawler

[PJ7] 2021.01-2021.12 (1 year), Challenges for Next-Generation Security R&D

[PJ6] 2021.03-2022.02 (1 year), A Fuzzing Seed Generation Technique Using Natural Language Processing Models

[PJ5] 2021.01-2023.05 (2 years 5 months), Development and Maintenance of a Discord Bot for SAG (SSG Algorithm Group)

[PJ4] 2020.11-2021.07 (9 months), Automated Building Fuzzing Environment Using Test Framework

[PJ3] 2020.09-2022.06 (1 year 10 months), Development of Security Primitives for Unmanned Vehicles

[PJ2] 2019.10-2019.11 (2 months), K-Cyber Security Challenge 2019: AI-based Network Threat Detection

[PJ1] 2019.03-2022.05 (3 years 3 months), The Study of Algorithms

HONORS and AWARDS

[A5] 2021. 4th Place (Excellence Prize). 2nd 2021 KOSPO Web Service Information Security Contest, KOSPO

[A4] 2019. Honorable Mention. ICPC SEOUL 2019, ICPC.Foundation

[A3] 2019. 3rd Place (정보화기획참모부장상). Cybersecurity Contest, Republic of Korea Army

[A2] 2019. 1st Place (Top Prize). Sejong Startup Idea Festival, Sejong University

[A1] 2017. 4th Place. SW Contest, Sejong University

PUBLICATION

PAPER

[P7] Jeong, S., **Ryu, M.**, Kang, H., and Huy Kang Kim. (2023). "Infotainment System Matters: Understanding the Impact and Implications of In-Vehicle Infotainment System Hacking with Automotive Grade Linux". *In Proceedings of the Thirteenth ACM Conference on Data and Application Security and Privacy (CODASPY '23)*, 201-212. <https://doi.org/10.1145/3577923.3583650>

[P6] Jeon, S., **Ryu, M.**, Kim, D., & Kim, H. K. (2022). "Automatically Seed Corpus and Fuzzing Executables Generation Using Test Framework". *IEEE Access*, 10, 90408–90428. <https://doi.org/10.1109/ACCESS.2022.3202005>

[P5] Kim, D., Jeon, S., **Ryu, M.**, & Kim, H. K. (2022). "A Fuzzing Seed Generation Technique Using Natural Language Processing Model". *Journal of the Korea Institute of Information Security & Cryptology*, 32(2), 417–437. <https://doi.org/10.13089/JKIISC.2022.32.2.417>

[P4] **Ryu, M.**, Kim, D. Y., Jeon, S., & Kim, H. K. (2021). "Automated Building Fuzzing Environment Using Test Framework". *Journal of the Korea Institute of Information Security & Cryptology*, 31(4), 587–604. <https://doi.org/10.13089/JKIISC.2021.31.4.587>

[P3] Cho, S., Lee, G., **Ryu, M.**, Jin, H., Kim, Y., Choi, S., Park, K. (2019). "PF-RNG: Proposal of Entropy Collection Model for Wireless Secure Communication Channel of Ultra Low-Cost Computing Device". *KING Spring Conference*, 202-205.

[P2] Kim, Y., Won, H., Jung, J., **Ryu, M.**, Kim, H., Kim, Y., Kim, K. (2018). "Study on packet extraction before encryption with SSL". *Conference on Information Security and Cryptography-Winter*.

[P1] Won, H., Kim, H., Jung, J., **Ryu, M.**, Kim, Y., Kim, Y., Kim, K. (2018). "Research for checking IM Security". *The Spring Conference of Korea Multimedia Society*.

PATENTS

[PT3] Seo, K., **Ryu, M.**, Kim, H., (2022). "Method for Recording event data of unmanned vehicle". KR Patent Application No. 10-2021-0165397

[PT2] Kim, H., Yoo, J., Jeong, S., Park, E., Seo, K., **Ryu, M.** (2022). "Generative adversarial network model and training method to generate message id sequence on unmanned moving objects". KR Patent Application No. 10-2021-0009487 & US Patent Application No. 17/532,226

[PT1] Kim, H., Yoo, J., Jeong, S., Park, E., Seo, K., **Ryu, M.** (2022). "Anomaly detection model using message id sequence on unmanned moving objects". KR Patent No. 10-2476359 & US Patent Application No. 17/532,272

DISCOVERED VULNERABILITIES (CVE)

[C4] CVE-2022-34096, <https://www.cve.org/CVERecord?id=CVE-2022-34096>, RESERVED.

[C3] CVE-2022-24597, <https://www.cve.org/CVERecord?id=CVE-2022-24597>, RESERVED.

[C2] CVE-2022-24596, <https://www.cve.org/CVERecord?id=CVE-2022-24596>, RESERVED.

[C1] CVE-2022-24595, <https://www.cve.org/CVERecord?id=CVE-2022-24595>, RESERVED.